TUNISIAN REPUBLIC
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH

UNIVERSITY OF TUNIS EL MANAR

NATIONAL ENGINEERING SCHOOL OF TUNIS

**Electrical Engineering Department**

# Worker's Internship Report


**Presented by :**

Nour KHEMIRI

**Class :**

2AGE2

**Supervised by :**

Mr. Hassen TOUTI


**Host company :**

NAIB BANK

مصرف شمال افريقيا الدولي

North Africa International Bank

Academic Year: 2024/2025

# Company data sheet

**Company name:** North African International Bank

**Address:** Avenue Kheireddine Pacha Ennassim Montplaisir

**Postal code:** 1002

**City:** Tunis

**Phone:** (+216) 71 950 800

**E-mail:** naib@naibank.com

**Web Site:** www.naibbank.com

**Number of employees:** 51-200

**Founded:** 1984

**Industry:** Banking

# Acknowledgements

First, I would like to express my gratitude to the entire pedagogical team at the National Engineering School of Tunis (ENIT) for granting me the invaluable opportunity to undertake this internship. The knowledge and wisdom imparted by them over the past year laid the foundation for my growth and development.

I am also thankful to the entire North Africa International Bank team for placing their trust in me and allowing me to embark on this transformative internship. Their unwavering guidance and faith in my abilities have played a pivotal role in shaping my academic and professional progress.

I extend my sincere appreciation to Mr. Zoubair CHOUAIB, head of IT department, and Mr Hassen TOUTI, member of the IT team in the bank who have been instrumental in my internship experience. Their mentorship and support have not only enriched my learning but have also contributed significantly to the successful conclusion of this internship.

This opportunity has allowed me to hone my technical skills and put my academic knowledge to the test. I am committed to leveraging the expertise I have gained to continue working towards my career goals and aspirations.

# List of Acronyms

**HFSQL** HyperFileSQL

**IDE** Integrated Development Environment

**UI** User Interface

**PK** Primary Key

**FK** Foreign Key

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# General introduction

In today's fast-evolving banking and financial services sector, developing efficient, secure, and user-friendly applications has become increasingly important. Mobile banking solutions, in particular, play a crucial role in improving client access to financial services and streamlining internal administrative processes.

North Africa International Bank (NAIB), a growing financial institution, stands out for its commitment to innovation and customer-oriented solutions. Their focus on improving client services, managing company accounts, and automating routine banking tasks inspired my interest as an engineering student at ENIT, in mobile app development and database integration.

During my internship at NAIB, I contributed to the development of a mobile banking solution consisting of two Mobile applications: one for clients and another for administrators. This experience allowed me to develop practical skills in WinDev Mobile, WLanguage, and database management.

In this report, I will first introduce NAIB and all the tools used during this internship. Then, I will provide a detailed overview of the tasks I undertook during my internship, highlighting the design, development, and testing of the mobile applications.

This internship has strengthened my interest in software development for mobile and administrative applications, particularly in projects that require database integration, real-time communication features, and user-focused design.

# Chapter 1

# Company's presentation

## 1.1 North Africa International Bank

NAIB was established in Tunis in November 1984 as an offshore banking unit, with an authorized share capital of USD 30,000,000.00, equally subscribed between the Tunisian State and the Forebank.

The bank primarily focuses on international banking operations, having established a comprehensive international correspondent network and established business relations in major financial centers. Then, its favourable local context and its incentive offshore status helped to implement its own "business concept" targeted towards finalizing both resident and non-resident, small and medium-sized exporting enterprises 'foreign trade operations.

NAIB succeeded in this emerging sector and performed satisfactory financial results, enabling it to focus on wider market shares and on extending its international expertise towards foreign partnership and investment activities, which prospects for Tunisia, the base country, are more than attractive and promising.

Figure 1.1: Logo NAIB

## 1.2 Overview of the project

The purpose of my internship was to contribute to the development of a mobile banking solution that the company is building to enhance client services and internal management processes. The team needed additional support to accelerate the design and implementation of a prototype application, particularly in structuring its core features and ensuring smooth navigation between different modules. The project focused on creating two WinDev Mobile applications. At a high level, the goal was to produce a functional demo that illustrates how these apps could streamline operations, improve the user experience, and serve as a foundation for future integration with real banking data. The expected outcome was not only to deliver working prototypes but also to validate the technical feasibility of the solution and identify areas for further improvement before full-scale development.

### 1.2.1 Project Description

The internship focused on the design and development of a mobile banking solution composed of two WinDev Mobile applications:

- **Client App:** Allowing users to log in, view account balances, access recent transactions, and check the status or make new requests (Checkbooks, Debt).
- **Admin App:** Enabling administrators to manage company and user accounts, review financial data, and process operations or requests.

### 1.2.2 Functional Requirements

The applications were designed to meet the following high-level requirements:

- Secure login and sign-in flows with role-based access (Treasurer, employee, etc).
- Clear and responsive interfaces for displaying balances, transactions, and user data.
- Data consistency across both apps, ensuring that admin updates are reflected on the client side.
- Scalability to allow migration from test data to a fully operational HyperFileSQL (HFSQL) database.

## 1.3 Tools and Technologies Used

The environment consisted of a Windows workstation with WinDev Mobile installed, connected to Android and iOS test devices for real-time debugging. No external servers were required at this stage because the database was configured to run locally using HFSQL Classic.

### 1.3.1 WinDev Express

A free, feature-limited version of WinDev, an Integrated Development Environment (IDE) for creating Windows desktop applications. It includes visual design tools, built-in database connectivity, and support for the WLanguage programming language, making it suitable for learning and prototyping.

Figure 1.2: WinDev logo

### 1.3.2 WinDev Mobile Express

The mobile counterpart of WinDev Express, used to build Android and iOS applications. It allows rapid User Interface (UI) design, integration with local or remote databases, and native compilation for mobile devices. Like WinDev Express, it is free but includes some restrictions compared to the full edition.

Figure 1.3: WinDev Express logo

### 1.3.3 WLanguage

The native programming language used across WinDev, WinDev Mobile, and WebDev. It is designed to be simple, readable, and tightly integrated with the IDE, providing high-level commands for user interfaces, database access, and business logic without requiring extensive low-level coding.

Figure 1.4: WLanguage

### 1.3.4 HFSQL Classic

A lightweight, file-based database engine included with the WinDev suite. It stores data locally in .fic files and requires no dedicated server, making it ideal for embedded applications, prototypes, or small-scale deployments while remaining compatible with the more scalable HFSQL Client/Server edition.

Figure 1.5: HFSQL logo

Each data file can be manipulated by a WINDEV Mobile application. These manipulations are performed using HFSQL functions (Hxxx functions).

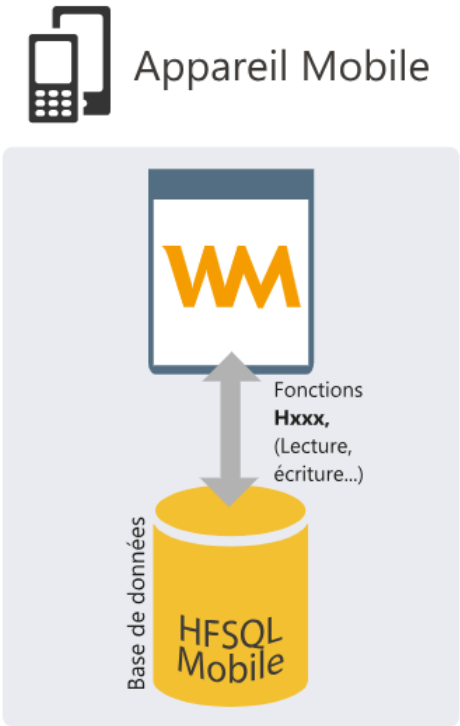



Figure 1.6: Manipulating an HFSQL Classic database from a mobile device

# Chapter 2

# Development Process

The development process began with an analysis of the requirements and a clear separation of responsibilities between the Client Application and the Admin Application. A basic workflow was defined to identify the essential features that needed to be implemented during the internship timeframe. The strategy was to first create functional navigation flows with hardcoded data, then progressively integrate the database to ensure data persistence and consistency.

## 2.1 User Interface Design

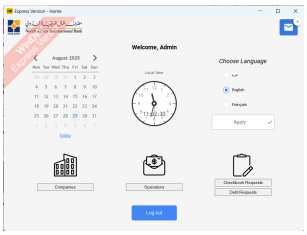
The applications were developed using the WinDev IDE, which provides drag-and-drop tools for designing windows and interface elements.

In the **Client App**, the focus was on creating simple and intuitive screens where users could log in, view balances, check recent transactions, and check the status or make new requests (Checkbooks, Debt).

In the **Admin App**, the design emphasized management features such as browsing companies, viewing and verifying associated users, handling operations and requests.

Screens were built incrementally to allow early testing of navigation and usability.

Client App Home Screen




Admin App Home Screen

Figure 2.1: Home Screens

## 2.2 Database Creation and Integration

Initially, the applications worked with hardcoded data to validate the navigation between windows and ensure correct display of information. This made it possible to test workflows without delays caused by database setup. Once the main flows were functional, the next step was to design an HFSQL Classic database, that was then linked to the Client App, enabling real sign-in and login operations. This transition from static data to a database-driven approach was an important milestone in validating the scalability of the project.

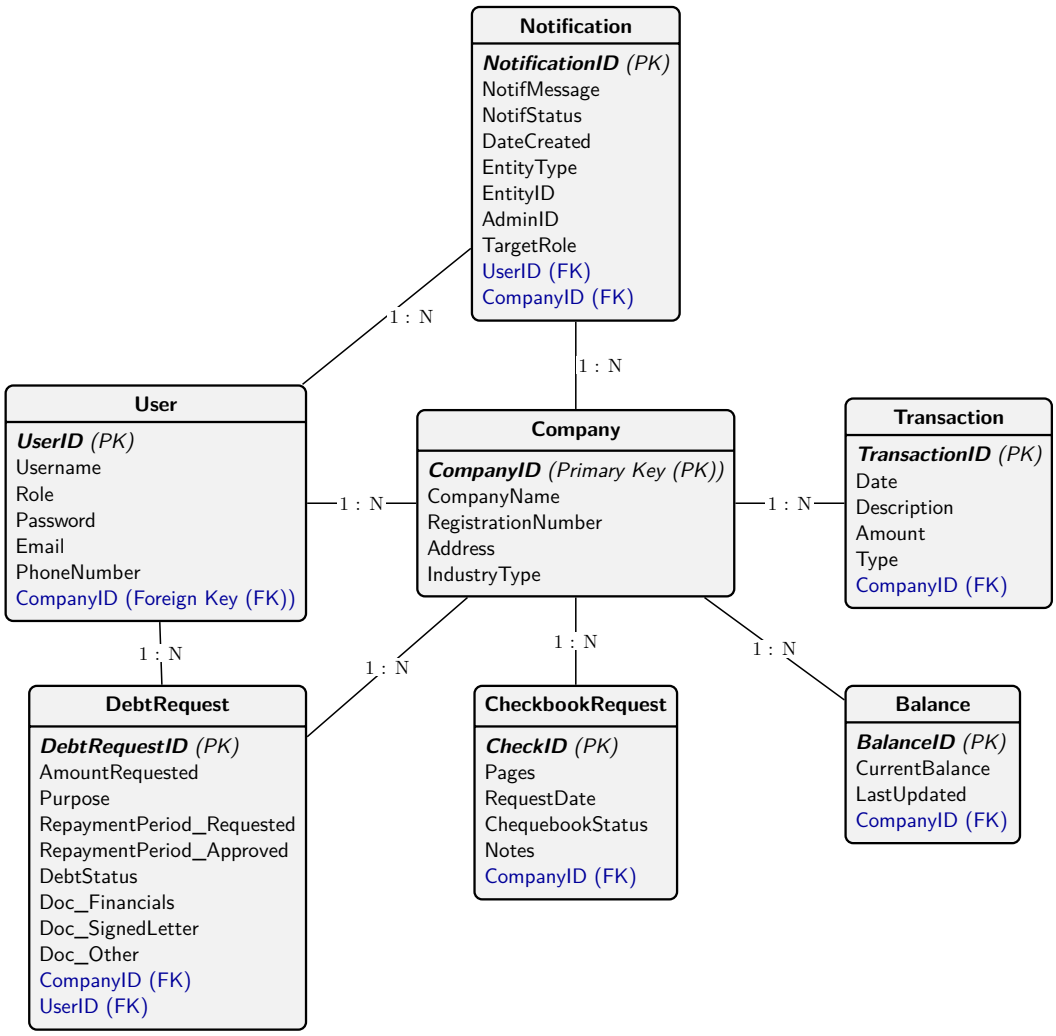



Figure 2.2: ER Diagram for the HFSQL Database

## Company

| Field | Type | Description |
|---|---|---|
| CompanyID (PK) | Integer | Unique company identifier |
| CompanyName | Varchar(100) | Company name |
| RegistrationNumber | Varchar(50) | Legal registration number |
| Address | Varchar(150) | Address |
| IndustryType | Varchar(50) | Industry sector |

## User

| Field | Type | Description |
|---|---|---|
| UserID (PK) | Integer | Unique user identifier |
| Username | Varchar(50) | Login/visible name |
| Role | Varchar(30) | e.g., Treasurer |
| Password | Varchar(100) | Password hash |
| Email | Varchar(120) | Email address |
| PhoneNumber | Varchar(20) | Phone number |
| CompanyID (FK) | Integer | References *Company.CompanyID* |

## Transaction

| | | |
|---|---|---|
| TransactionID (PK) | Integer | Unique transaction identifier |
| Date | Date | Transaction date |
| Description | Varchar(150) | Notes/details |
| Amount | Currency | Amount (+/) |
| Type | Varchar(20) | Credit/Debit |
| CompanyID (FK) | Integer | References *Company.CompanyID* |

## Balance

| Field | Type | Description |
|---|---|---|
| BalanceID (PK) | Integer | Row identifier |
| CurrentBalance | Currency | Current balance |
| LastUpdated | Date | Timestamp of snapshot |
| CompanyID (FK) | Integer | References *Company.CompanyID* |

## DebtRequest

| Field | Type | Description |
|---|---|---|
| DebtRequestID (PK) | Integer | Unique request ID |
| AmountRequested | Currency | Requested amount |
| Purpose | Varchar(120) | Purpose of the loan |
| RepaymentPeriod_Requested | Integer | Months to repay requested |
| RepaymentPeriod_Approved | Integer | Months to repay approved |
| DebtStatus | Varchar(20) | Pending/Approved/Rejected |
| Doc_Financials | Varchar(200) | Document reference/path |
| Doc_SignedLetter | Varchar(200) | Document reference/path |
| Doc_Other | Varchar(200) | Document reference/path |
| CompanyID (FK) | Integer | References *Company.CompanyID* |
| UserID (FK) | Integer | References *User.UserID* |

## CheckbookRequest

| Field | Type | Description |
|---|---|---|
| CheckID (PK) | Integer | Unique request ID |
| Pages | Integer | Number of pages |
| RequestDate | DateTime | Request date |
| ChequebookStatus | Varchar(20) | Pending/Issued/Rejected |
| Notes | Varchar(150) | Optional note |
| CompanyID (FK) | Integer | References *Company.CompanyID* |

## Notification

| Field | Type | Description |
|---|---|---|
| NotificationID (PK) | Integer | Unique notification ID |
| NotifMessage | Varchar(200) | Message content |
| NotifStatus | Varchar(20) | Unread/Read/Archived |
| DateCreated | DateTime | Creation time |
| EntityType | Varchar(30) | Target entity (e.g., Transaction) |
| EntityID | Integer | Target entity ID |
| AdminID | Integer | Admin ID |
| TargetRole | Varchar(200) | Admin or User |
| UserID (FK) | Integer | References *User.UserID* |

## 2.3 Technical Contributions

Throughout the internship, several windows and features were designed and implemented.

### 2.3.1 Client App:

- **WIN_Sign_In & WIN_Sign_In2** : two-step process to register a new company and its first user. The first collects company data, the second sets user credentials and role.
- **WIN_Balance** : shows the user's current account balance clearly.
- **WIN_Transactions** : lists recent deposits, withdrawals, and transfers for easy tracking.
- **WIN_Requests** : lets clients submit requests (checkbooks, debts), flagged for admin review.

### 2.3.2 Admin App:

- **WIN_Companies** : lists registered companies with details and users for easy management.
- **WIN_Operations** : shows balances and transactions by company, giving admins quick financial overviews.
- **WIN_DebtRequest** : displays client debt requests by company for admin review and approval.
- **WIN_CheckbookRequest** : shows checkbook requests by company, helping admins track and process them.

### 2.3.3 Both Apps:

- **WIN_Login** : secure login validating user credentials against the HFSQL database.
- **WIN_Home** : main dashboard with quick navigation and notifications.
- **WIN_Notification** : enables real-time communication between Client and Admin apps. A counter shows pending notifications, and tapping one redirects users to the related window (e.g., requests or transactions).
- **Multi-language Support** : full support for Arabic, English, and French. Users can switch language, with all windows and controls updating dynamically for better usability.

## 2.4 Challenges and Solutions

Some challenges were encountered during the project:

- **Database connection issues** : Early attempts to link the HFSQL database resulted in errors due to incorrect path settings. This was solved by configuring the database to use relative paths and ensuring `.fic` files were correctly initialized.
- **Navigation consistency** : Some windows did not return correctly to the home screen, which created loops or dead ends. The problem was addressed by standardizing navigation buttons and using a consistent back-to-home logic.
- **Time constraints** : Given the limited internship period, not all planned features could be fully implemented. To manage this, the focus was kept on delivering functional prototypes of core features first, leaving advanced functions for future development.
- **Learning Curve with WinDev** : As this was my first time using WinDev and HFSQL, I had to learn everything from scratch in a short time. The somewhat outdated tools made it harder to find resources, but by focusing on core concepts and experimenting, I managed to build functional apps within the internship period.

The development process demonstrated a full cycle from planning, UI design, and initial prototyping with hardcoded data to functional integration with an HFSQL Classic database. Despite challenges, the applications achieved the objective of delivering a working prototype that can serve as a foundation for future iterations.

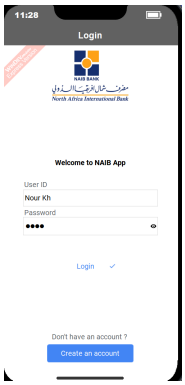
# Chapter 3

# Results and Testing

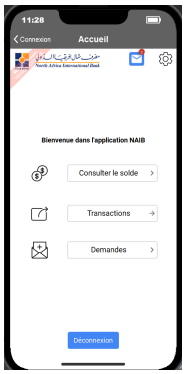
The internship resulted in two functional prototype applications.

## 3.1 Functional Demo

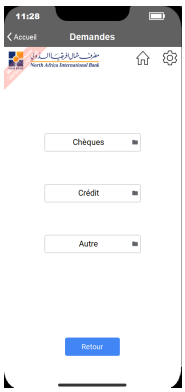
This is an example of a checkbook request by a Client :



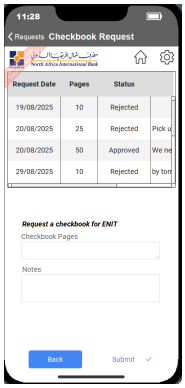

(a) WIN_Login



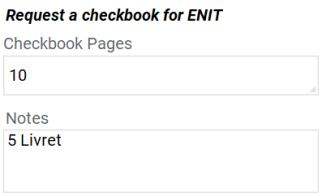

(b) WIN_Home



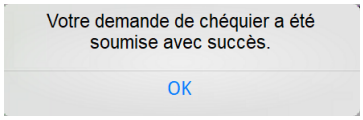

(c) WIN_Requests




(d) WIN_Checkbook



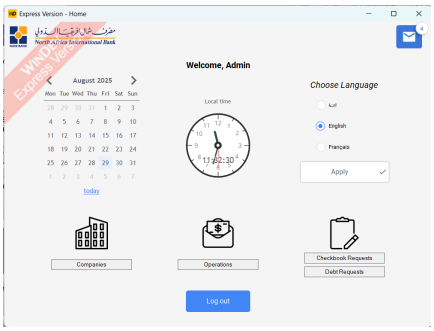

(e) Checkbook request



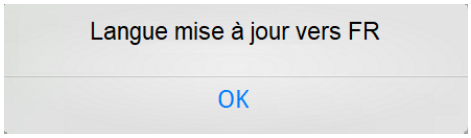

(f) Checkbook submission success

Figure 3.1: Example of checkbook request from the Client App demo

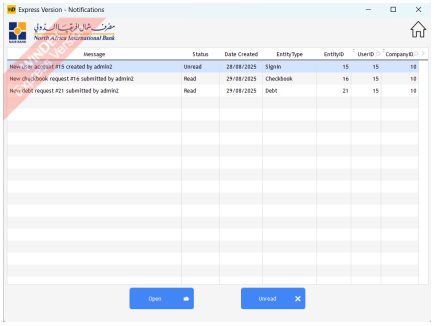
This is an example of a language change and a checkbook request review by an admin :



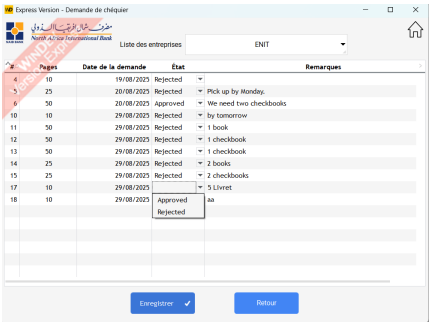

(a) WIN_Home




(b) Language_change




(c) WIN_Notification




(d) Checkbook requests displayed by company

Figure 3.2: Example of language change and checkbook review from the Admin App demo

## 3.2 Testing and Validation

- Each window was tested step by step to ensure buttons, inputs, and database operations worked correctly.
- Hardcoded data was first used to verify navigation and calculations before switching to HFSQL integration.
- Test scenarios included login with valid/invalid credentials, submitting requests, checking balances, and verifying that notifications appeared in both apps.

## 3.3 Supervisor Feedback

- Supervisors confirmed that the apps demonstrated the intended features within the short internship timeline.
- The smooth notification system and multi-language support were highlighted as particularly useful.

# Conclusion

During my internship at NAIB, I realized the significant potential of the project, particularly in developing mobile applications that seamlessly connect clients and administrators. The groundwork established during this internship can serve as the foundation for more advanced applications, including database enhancements, user interface refinements, and integration with real banking data to create a fully functional system.

This experience has greatly enriched my understanding of mobile application development, database management, and real-time notification systems, reinforcing my passion for software development and IT solutions. I am incredibly grateful to my supervisor and the team at NAIB for their invaluable guidance, support, and shared expertise, which have had a lasting impact on my professional growth.

Working during this internship and gaining insights into the processes has been a unique opportunity that I highly value. It has allowed me to develop practical skills and navigate the collaborative nature of a professional environment.

This report highlights the knowledge and technical skills I acquired in this short but enriching experience. I'm confident that this journey will propel my academic and professional progress in the field.

# Bibliography


[1] PC SOFT, *Discover windev mobile*, https://help.windev.com/en-US/?1410087562&name=discover_windev_mobile, Accessed: 01/08/2025.

[2] PC SOFT, *My first android/ios project*, https://help.windev.com/en-US/?1410087563&name=1_first_androidios_project, Accessed: 01/08/2025.

[3] PC SOFT, *Wlanguage: A simple and powerful language*, https://help.windev.com/en-US/?1410087570&name=1_wlanguage, Accessed: 01/08/2025.

[4] PC SOFT, *Deploying an android application*, https://help.windev.com/en-US/?1410087576&name=deploying_android_application, Accessed: 01/08/2025.

[5] PC SOFT, *Deploying an ios application*, https://help.windev.com/en-US/?1410087519&name=deploying_ios_application, Accessed: 01/08/2025.

[6] NAIB, *Official website*, http://www.naibbank.com/, Accessed: 01/08/2025.

[7] PC SOFT, *Hfsql classic*, https://doc.pcsoft.fr/?1410086918&name=4_hfsql_classic, Accessed: 06/08/2025.

[8] PC SOFT, *Push notification*, https://help.windev.com/en-us/?1000021015&name=push_notification, Accessed: 27/08/2025.